**This assignment will not be graded and is only for practice.**

## Level 1

1) If $||a|| = 3$, $||b|| = 2\sqrt{2}$ and angle between $a$ and $b$ is $45^\circ$, then the value of $a.b$ is

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 6

***1 point***

2) Given that $\|a\| = 3$ and $\|b\| = 5$ and $a \cdot b = 7.5$, for two vectors $a$ and $b$ in $\mathrm{R}^n$. Find the angle (in degrees) between the two vectors $a, b$. Note: $\cos 60^\circ = 1/2$, $\cos 30^\circ = \sqrt{3}/2$. ***1 point***

- ○ $30^\circ$
- ○ $60^\circ$
- ○ $120^\circ$
- ○ $150^\circ$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$60^\circ$

3) Consider two vectors $a = (3, 4, -1)$, $b = (2, -1, 2)$ in $\mathrm{R}^3$. Choose the correct options. ***1 point***

- ☐ Length/Norm of $a$ is $\sqrt{26}$ and that of $b$ is $3$.
- ☐ Length/Norm of $a$ is $5$ and that of $b$ is $3$.
- ☐ Length/Norm of $a$ is $\sqrt{24}$ and that of $b$ is $\sqrt{7}$.
- ☐ Angle between the two vectors is $90$ degrees.
- ☐ Angle between the two vectors is approximately $0$ degrees.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Length/Norm of $a$ is $\sqrt{26}$ and that of $b$ is $3$.
Angle between the two vectors is $90$ degrees.

4) If the dot product of two vectors is zero, then choose the correct option. ***1 point***

- The two vectors are perpendicular to each other.
- The two vectors are parallel to each other.
- The two vectors are exactly the same.
- The length of the two vectors must be the same.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The two vectors are perpendicular to each other.

**Level 2**

5) If $a = (6, -1, 3), b = (4, c, 2)$ where $a, b \in \mathrm{R}^3$ and $a$ and $b$ are perpendicular to each other, then find the value of $c$?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 30

***1 point***

6) Consider two vectors $a = (1, 2), b = (2, 2)$ and $\theta$ is the angle between them. The sum of the two vectors is given by $c = a + b$. Choose the correct options. ***1 point***

- [ ] Length/Norm of $c$ is $5$.
- [ ] Length/Norm of $c$ is $25$.
- [ ] Length/Norm of $c$ is $4$.
- [ ] $\cos\theta = \frac{3}{\sqrt{10}}$.
- [ ] $\cos\theta = \frac{3}{\sqrt{5}}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Length/Norm of $c$ is $5$.
$\cos\theta = \frac{3}{\sqrt{10}}$.

7) Consider $3$ vectors $a, b, c$ in $\mathrm{R}^3$ and a scalar $\lambda$ in $\mathrm{R}$. Choose the set of correct options. Note: $(.)$ represents the dot product. ***1 point***

- [ ] $\lambda(a \cdot b) = (\lambda a) \cdot b$
- [ ] $\lambda(a \cdot b) = (\lambda b) \cdot a$
- [ ] $\lambda(a \cdot b) = (\lambda a) \cdot (\lambda b)$

- [ ] $(a+c)\cdot b = a\cdot b + c\cdot b$
- [ ] $(a+c)\cdot b = a\cdot c + c\cdot b$
- [ ] $a\cdot a = 0, b\cdot b = 0$ if and only if $a, b$ are null vectors.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\lambda(a\cdot b) = (\lambda a)\cdot b$
$\lambda(a\cdot b) = (\lambda b)\cdot a$
$(a+c)\cdot b = a\cdot b + c\cdot b$
$a\cdot a = 0, b\cdot b = 0$ if and only if $a, b$ are null vectors.

8) Consider two vectors $a = (1, 3, 5, 7, 9)$ and $b = (2, 4, 6, 8, 10)$. Choose the set of correct options. ***1 point***

- [ ] The length of $b$ is more than $a$.
- [ ] The length of $a$ is more than $b$.
- [ ] The length of $(a-b)$ is $\sqrt{5}$.
- [ ] The length of $(a-b)$ is $\sqrt{50}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The length of $b$ is more than $a$.
The length of $(a-b)$ is $\sqrt{5}$.

Your score is: 0/8